SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Design Task

Code of Conduct:

I Nalluri Akhilesh Kumar(192365074) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Nalluri Akhilesh Kumar
Signature of the student:

Design Task

1. **Hybrid AI Recommendation Engine Design**

   Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

2. **Smart Disaster Detection System**

   Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

3. **AI-Based Fraud Detection Framework**

   Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

4. **Autonomous Supply Chain Optimization System**

   Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

5. **Personalized Learning Analytics Platform**

   Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration
Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined

Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation

QUESTION 1

HYBRID AI RECOMMENDATION ENGINE DESIGN

Introduction

Recommendation systems play a vital role in modern digital platforms such as Amazon, Netflix, Spotify, and YouTube. Their primary objective is to suggest products, movies, songs, or content that match user preferences. Traditional recommendation methods often struggle with scalability and personalization when handling millions of users and items. A hybrid recommendation engine combines multiple recommendation approaches to overcome these limitations.

The proposed hybrid recommendation system integrates content-based filtering and collaborative filtering techniques using PySpark RDDs. Content-based filtering recommends items similar to those previously liked by users, while collaborative filtering identifies similarities among users and predicts preferences based on collective behavior.

Objectives

1. Develop a scalable recommendation framework.
2. Improve recommendation accuracy.
3. Process large-scale user-item interactions.
4. Adapt recommendations in real time.
5. Enhance user satisfaction.

System Architecture

The system consists of five layers:

Data Collection Layer

This layer gathers:

- User profiles
- Product information

- User ratings
- Search history
- Clickstream data

Distributed Processing Layer

PySpark RDDs partition data across multiple cluster nodes. Distributed processing enables:

- Parallel computation
- Fault tolerance
- Faster recommendation generation

Content-Based Filtering Module

The module analyzes:

- Item descriptions
- Categories
- Features
- User interests

Similarity measures are calculated using cosine similarity and TF-IDF techniques.

Collaborative Filtering Module

This module identifies:

- Similar users
- Similar items
- Hidden preference patterns

Matrix factorization and neighborhood methods are employed.

Intelligent Agent Layer

Multiple intelligent agents continuously monitor user activities and improve recommendation quality.

Role of PySpark RDDs

RDD transformations include:

- map()
- filter()
- flatMap()
- join()
- reduceByKey()

RDD actions include:

- collect()
- count()
- take()
- saveAsTextFile()

The distributed architecture allows processing of billions of user-item interactions efficiently.

Intelligent Agent Functions

User Behavior Agent

Tracks:

- Browsing history
- Purchase behavior
- Search patterns

Recommendation Agent

Generates personalized suggestions.

Feedback Agent

Analyzes ratings and reviews.

Learning Agent

Updates recommendation models dynamically.

Advantages

- High scalability
- Real-time recommendations
- Improved personalization
- Better customer engagement
- Reduced cold-start issues

Applications

- E-commerce
- Online streaming
- Social networking
- Digital advertising

Future Enhancements

- Deep learning integration
- Context-aware recommendations
- Voice-based recommendation systems

Conclusion

The hybrid recommendation engine combines distributed computing and intelligent agents to deliver scalable and adaptive recommendations for large-scale applications.

QUESTION 2

SMART DISASTER DETECTION SYSTEM

Introduction

Natural disasters such as floods, earthquakes, landslides, and cyclones cause significant economic losses and human casualties. Early detection and timely response are essential to minimize disaster impacts. Artificial Intelligence and distributed computing provide powerful tools for disaster monitoring and prediction.

The proposed Smart Disaster Detection System uses PySpark RDDs to process data collected from multiple sources including satellites, sensors, weather stations, and social media platforms.

Objectives

1. Detect disasters early.
2. Fuse multi-source data.
3. Generate real-time alerts.
4. Support emergency management.

System Architecture

Data Acquisition Layer

Data sources include:

- Satellite images
- Seismic sensors
- Water-level sensors
- Weather stations
- Social media feeds

Distributed Processing Layer

RDDs distribute incoming data among multiple nodes for parallel processing.

AI Analytics Layer

Machine learning models identify:

- Flood risks
- Earthquake patterns
- Cyclone development

Agent Coordination Layer

Multiple intelligent agents collaborate to assess risks.

Data Fusion Techniques

Feature-Level Fusion

Combines raw sensor measurements.

Decision-Level Fusion

Combines outputs from multiple AI models.

Hybrid Fusion

Integrates feature and decision-level fusion.

Intelligent Agents

Monitoring Agent

Continuously gathers environmental data.

Risk Assessment Agent

Calculates disaster probability.

Alert Agent

Generates warning messages.

Response Agent

Coordinates emergency actions.

Advantages

- Faster disaster detection
- Improved situational awareness
- Reduced response time
- Enhanced public safety

Applications

- Flood monitoring
- Earthquake prediction
- Wildfire detection
- Tsunami warning systems

Conclusion

Distributed AI systems significantly improve disaster prediction accuracy and emergency response efficiency.

QUESTION 3

AI-BASED FRAUD DETECTION FRAMEWORK

Introduction

Financial fraud causes billions of dollars in losses annually. Traditional fraud detection approaches often fail to identify sophisticated attack patterns. AI and distributed computing enable real-time fraud detection using large-scale transaction data.

Objectives

1. Detect suspicious transactions.
2. Reduce false positives.
3. Improve fraud detection accuracy.
4. Enable continuous learning.

System Architecture

Transaction Collection Layer

Collects:

- Credit card transactions
- Online banking transactions
- Digital payment records

Distributed Processing Layer

RDDs process millions of transactions simultaneously.

Anomaly Detection Engine

Identifies abnormal transaction behavior.

Alert Management Layer

Generates alerts and notifications.

Distributed Anomaly Detection

Algorithms include:

- Isolation Forest
- K-Means Clustering
- Local Outlier Factor
- Autoencoders

RDD operations perform distributed feature extraction and analysis.

Intelligent Agents

Monitoring Agent

Tracks transactions.

Detection Agent

Identifies anomalies.

Alert Agent

Notifies security teams.

Learning Agent

Updates fraud patterns automatically.

Advantages

- Real-time processing
- High scalability
- Better accuracy
- Reduced financial losses

Applications

- Banking
- Insurance
- E-commerce
- Digital payments

Conclusion

AI-powered fraud detection frameworks provide efficient and scalable protection against financial fraud.

QUESTION 4

AUTONOMOUS SUPPLY CHAIN OPTIMIZATION SYSTEM

Introduction

Supply chain management involves inventory control, logistics planning, demand forecasting, and supplier management. AI-driven distributed systems improve operational efficiency and reduce costs.

Objectives

1. Optimize inventory.
2. Improve logistics planning.
3. Forecast demand accurately.
4. Reduce operational costs.

System Architecture

Inventory Module

Tracks stock levels.

Forecasting Module

Predicts future demand.

Logistics Module

Optimizes transportation routes.

Agent Coordination Layer

Enables decentralized decision-making.

PySpark RDD Usage

RDDs process:

- Sales records
- Supplier information
- Transportation data
- Warehouse records

Transformations:

- map()
- filter()
- groupByKey()
- reduceByKey()

Intelligent Agents

Inventory Agent

Maintains stock levels.

Forecasting Agent

Predicts market demand.

Logistics Agent

Optimizes delivery routes.

Supplier Agent

Evaluates supplier performance.

Advantages

- Cost reduction
- Faster deliveries
- Better inventory utilization
- Improved customer satisfaction

Applications

- Retail
- Manufacturing
- Healthcare logistics
- E-commerce

Conclusion

Autonomous supply chain systems improve efficiency through distributed processing and intelligent agent collaboration.

QUESTION 5

PERSONALIZED LEARNING ANALYTICS PLATFORM

Introduction

Educational institutions generate enormous amounts of learning data. AI-based analytics platforms help educators understand student behavior and personalize learning experiences.

Objectives

1. Monitor student performance.
2. Personalize learning paths.
3. Provide real-time feedback.
4. Improve learning outcomes.

System Architecture

Data Collection Layer

Collects:

- Quiz scores
- Attendance records
- Assignment submissions
- Learning activity logs

Distributed Analytics Layer

PySpark RDDs process educational data at scale.

AI Recommendation Layer

Generates personalized learning recommendations.

Intelligent Tutoring Layer

Provides adaptive guidance.

Intelligent Agents

Learning Agent

Tracks progress.

Recommendation Agent

Suggests learning resources.

Feedback Agent

Provides personalized feedback.

Assessment Agent

Evaluates performance.

Adaptive Learning Features

- Dynamic content delivery
- Personalized assignments
- Intelligent assessments
- Progress monitoring

Advantages

- Enhanced learning outcomes
- Increased engagement
- Personalized education
- Scalable analytics

Applications

- Universities
- Online learning platforms
- Corporate training systems
- Skill development programs

Future Enhancements

- Virtual tutors
- AI-powered mentoring
- Learning analytics dashboards

Conclusion

The Personalized Learning Analytics Platform uses distributed AI and intelligent agents to deliver scalable and adaptive educational experiences.

OVERALL CONCLUSION

Distributed Artificial Intelligence systems built using PySpark RDDs and intelligent agents provide scalable solutions for modern real-world challenges. Recommendation systems improve personalization, disaster detection systems enhance public safety, fraud detection frameworks protect financial assets, supply chain optimization improves operational efficiency, and learning analytics platforms transform education. The integration of distributed processing and intelligent agents enables real-time decision-making, continuous learning, and high-performance analytics, making these systems essential for future intelligent applications.